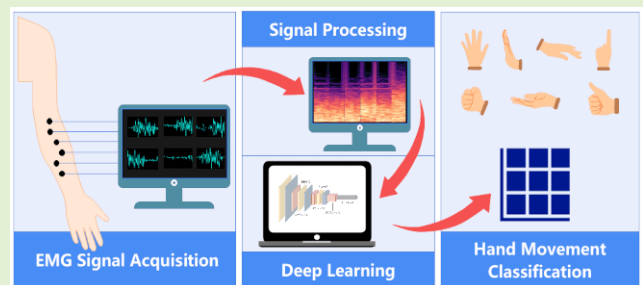


Spectral Image-Based Multiday Surface Electromyography Classification of Hand Motions Using CNN for Human–Computer Interaction

Muhammad Farrukh Qureshi^{ID}, Zohaib Mushtaq, Muhammad Zia ur Rehman, and Ernest Nlandu Kamavuako^{ID}

Abstract—Physiological signals such as electromyography (EMG) have been used in human–computer interaction (HCI) for medical applications. Wearable prostheses, such as robotic limbs, have seen a surge in popularity because of technological advancements in myoelectric interfaces. In spite of encouraging achievements with pattern-recognition-based control systems, user acceptability of prosthetic hands still needs improvement in control robustness. The purpose of this research is to compare multiday surface EMG (sEMG) recordings and measure the performance of convolutional neural network (CNN) to enhance myoelectric control. The performance metrics used in this study are accuracy, macro weighted precision (MWP), macro weighted recall (MWR), and macro $F1$ -score for eight able-bodied (healthy and nondisabled) and four amputee subjects. Using Mel-spectrogram-based sEMG data from both the able-bodied (healthy and nondisabled) and amputee participants, our proposed CNN has achieved a mean classification accuracy of $99.42\% \pm 0.42\%$ and $98.00\% \pm 0.58\%$ for the able-bodied (healthy and nondisabled) and amputee subjects for the within-day analysis, respectively. The proposed CNN outperformed other classifiers ($p \leq 0.05$) in the between-day analysis for twofold ($65.88\% \pm 10.1\%$ and $58.37\% \pm 9.11\%$) and for sevenfold validation ($88.73\% \pm 1.43\%$ and $77.35\% \pm 2.72\%$) using sEMG recordings from the able-bodied (healthy and nondisabled) and amputee subjects, respectively. The proposed CNN is compared with the pretrained transfer learning (TL) models and has achieved higher accuracy with lower computational cost. The results demonstrate that CNN can considerably increase the effectiveness of the pattern recognition myoelectric control schemes and can extract deep information from EMG data.

Index Terms—Deep convolutional neural network (CNN), electromyography (EMG), Mel spectrogram, pretrained transfer learning (TL).



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I. INTRODUCTION

HUMAN–COMPUTER interaction (HCI) applications have drawn increasing interest in myoelectric control devices in the past several decades [1], [2], [3]. The use of hand movements to control peripheral devices is a characteristic shared by these systems. Electromyography (EMG) has shown improved control and resilience compared with other approaches [4], [5]. To classify EMG signals, several researchers have used classical machine learning techniques such as artificial neural networks [6], support vector machines [7], [8], [9], [10], linear discriminant analysis [11], [12], and K -nearest neighbor [13], [14] for feature extraction. However, feature extraction is time-consuming and requires professional expertise to find the optimal feature set. Deep neural network (DNN) addresses this problem by creating a better representation from input data using multiple layers of neural networks.

TABLE I
ANALYSIS OF THE EXISTING STUDIES ON VARIOUS EMG DATASETS

Reference	Dataset	Subjects	Network	Accuracy	Categories	Paradigm
[19]	NinaPro	-	CNN	75%–84%	8	Intra-session
[20]	HD-sEMG acquired dataset	Healthy: 1	CNN	98.15%	6	Intra-session
[21]	NinaPro; BioPatRec; CapgMyo-DBa	-	CNN & RNN	82% - 99%	-	Intra-session
[22]	Ninapro & Myo Dataset	-	CNN	70% - 98.81%	7	Intra-session
[23]	NinaPro	Healthy: 67; Amputees 11	CNN	38.09% - 60.27%	50	Intra-session
[24]	CapgMyo-DBa & CSL-HDEMG	-	CNN	61.7% - 98.6%	27	Intra-session
[25]	Myo-Armband acquired dataset	Healthy: 35	GRU	77.85%	6	Inter-subject
[26]	NinaPro	-	SVM & CNN	66.9% - 68.23%	-	Inter-session
[27]	NinaPro & BioPatRec	-	CNN	B80% - 94%	-	Inter-subject
[28]	Self-Made Dataset	Healthy: 8	CNN	68%	10	Inter-subject / Inter-session
[29]	ISRMyo	Healthy: 6	CNN	80.25%	13	Inter-subject / Inter-session
[30]	CapgMyo-DB & NinaPro	Healthy: 50	RNN	34.8% - 95%	-	Inter-subject / Inter-session
[31]	CapgMyo & NinaPro & CSL-HDEMG	Healthy: 23	CNN	87.7% - 96.8%	-	Inter-subject / Inter-session

DNN has been used in image classification [28], speech recognition [29], sound classification [30], and video classification [31]. One of the most famous DNN is the convolutional neural network (CNN). In recent years, CNN-based surface EMG (sEMG) feature extraction improved hand gesture recognition for short-term and long-term classification. The CNN-based short-term classification (intrasession) has been extensively researched. Ding *et al.* [15] proposed a parallel multiscale convolution architecture with bigger kernel filter sizes. Tam *et al.* [16] demonstrated a CNN for real-time fine gesture detection for myoelectric control on one healthy individual with 98.15% accuracy. In [17], a hybrid architecture of the recurrent neural network (RNN) and CNN was developed and the achieved accuracy was 87.2%, 94.1%, 99.7%, and 94.5%. Chen *et al.* [18] evaluated a compact CNN model using NinaPro DB5 and Myo dataset with 70%–98.81% accuracy. Atzori *et al.* [19] assessed an average of 50 hand gestures in 67 able-bodied (healthy and nondisabled) individuals and 11 transradial amputees using CNN and obtained a classification accuracy between 38.09% and 60.27%. In [20], a 3-D CNN with 3-D kernels was used to gather the spatial and temporal characteristics from successive sEMG images for an HD-sEMG-based gesture recognition system and achieved 61.7%–98.6% accuracy. Nasri *et al.* [21] proposed Myo Armband gesture recognition. They used a gated recurrent unit (GRU) network to train on raw sEMG data and obtained 77.85% accuracy.

However, some works still provide results on long-term classification (intersession/intersubject). Zhai *et al.* [22] used a self-recalibrating CNN that does not need user retraining. The method was tested on 40 healthy and 11 amputee hands in the NinaPro database. Wei *et al.* [23] developed two-stream CNN based on sEMG features for NinaPro and BioPatRec with 94% and 80%–90% accuracy, respectively. Another CNN-based feature extraction technique was developed in [24] with 68% accuracy. In [25], a compact CNN architecture was developed and it achieved an accuracy of 80.25%. A two-stage RNN was deployed that achieved an accuracy between 34.8% and 95% in [26]. In [27], a CNN was applied and it achieved a classification accuracy of 87.7%–96.8%. Table I provides the summary of these studies.

Several factors contribute to data variability in EMG such as drift in sensor, sensor placement, and skin conductivity [26]. A long-term classification is necessary to mitigate this issue. As seen in Table I, few of these research studies

use long-term classification. Therefore, this study incorporates multiday data from [32] obtained from both healthy and amputee patients. Furthermore, since a raw EMG signal is 1-D, stochastically distributed, and lacks spectral information, it should be converted into a 2-D spectrogram before being fed into a CNN. When input to CNN, a spectrogram provides a better interpretation of spectrum information and demonstrates improved processing. The following are the primary and significant contributions of this experimental study.

- 1) Unique approach of Mel spectrogram images for classification of sEMG signals for upper limb is used.
- 2) Analysis of a proposed CNN on intrasession and intersession on the dataset for both the able-bodied (healthy and nondisabled) and amputee subjects is done.
- 3) The performance of the proposed CNN is compared with that of stacked sparse autoencoders (SSAEs) on the given dataset in [32].
- 4) The same proposed CNN is compared with well-known pretrained transfer learning (TL) weights such as AlexNet, ResNet, and DenseNet.

The rest of this article is structured as follows. In Section II, an introduction to the Mel spectrogram, preprocessing techniques, and the architecture of the proposed CNN are presented. The experimental results are presented in Section III. Section IV provides further discussion of the results, and Section V concludes this article.

II. METHODOLOGY

The generalized block diagram is illustrated in Fig. 1. The details of these blocks will be discussed in Sections II-C and II-D.

A. Mel-Spectrogram-Based Feature Extraction Technique

Spectrogram is a Fourier transform of raw signal and displays frequency content as a function of time and is widely used for signal visualization. The Mel spectrogram is a spectrogram that is divided into multiple points, each of which distributes frequencies and times evenly on a Mel frequency scale. In [30], the relationship is defined as

$$\text{Mel} = 2595 \times \log_{10} \left(1 + \frac{f}{700} \right) \quad (1)$$

and its inverse is given as

$$f = 700 \times \left(10^{\frac{\text{Mel}}{2595}} - 1 \right) \quad (2)$$

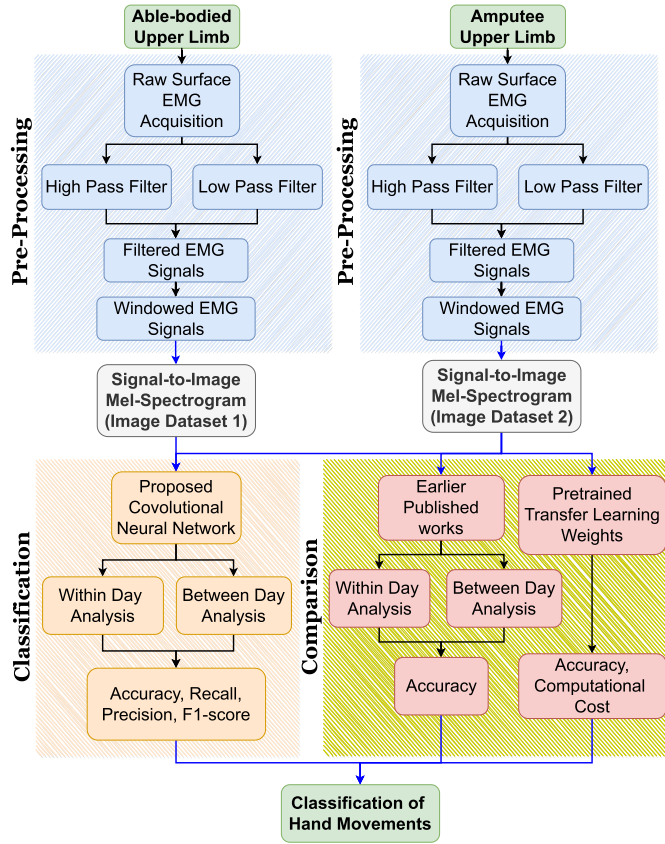


Fig. 1. Generalized block diagram of the work followed in this study.

where Mel is a Mel-based frequency, while the normal frequency is represented by f .

B. Experimental Datasets and Setup Description

In this article, a dataset from a previous work [32] is being used. The data used included eight able-bodied (healthy and nondisabled) volunteers and four transradial amputees. Table II contains the demographic information on the subjects. Three electrodes were connected with flexor carpi radialis, palmaris longus muscle, and flexor digitorum superficialis, while other three were connected with extensor carpi radialis longus, extensor digitorum, and extensor carpi ulnaris and provided sEMG data as six channels. Each participant performed 11 hand motions: hand open, hand close, supination, pronation, fine grasp, side grip, flex hand, extend hand, agree, pointer, and rest. Each subject performed seven sessions separated by 24 h. Each hand movement was repeated four times every session, with a 5-s contraction and relaxation period. Fig. 2 illustrates a recording of a single sEMG movement.

C. Preprocessing Technique

An 8-kHz sample rate is used to obtain data from sEMG signals. The signals are filtered at 10–500 Hz bandpass filter and then 50 Hz notch filter. Each raw signal is divided into 200 ms windowed signals with an overlapping increment of 29 ms as shown in Fig. 2. There are 11 movements with four repetitions; as a result, there are 400 windowed signals for

TABLE II
DETAILS OF THE USED DATASET

Type of Subject	Able-bodied	Amputee
No. of Subjects	8	4
Mean Age $\pm$ SD (years)	24.5 $\pm$ 0.22	34.8 $\pm$ 0.32
Gender		Male
Days		7
Movements		11
Number of Channels		6
Sampling frequency		8 kHz

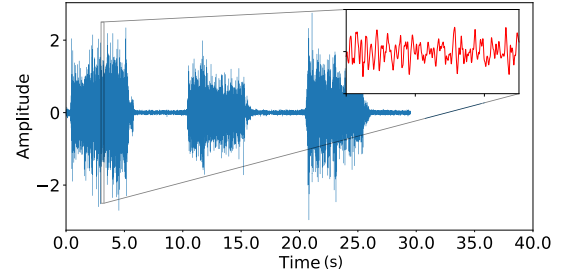


Fig. 2. Raw EMG single movement with four repetitions. The red signal is the windowed signal with a length of 200 ms.

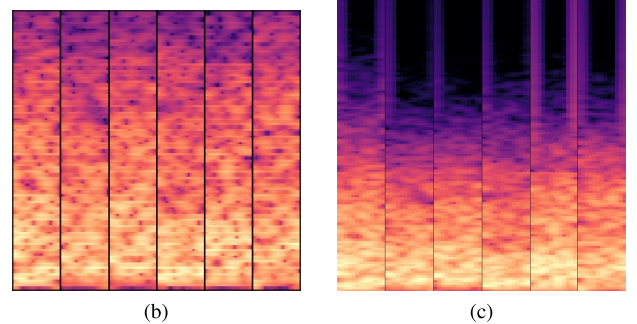
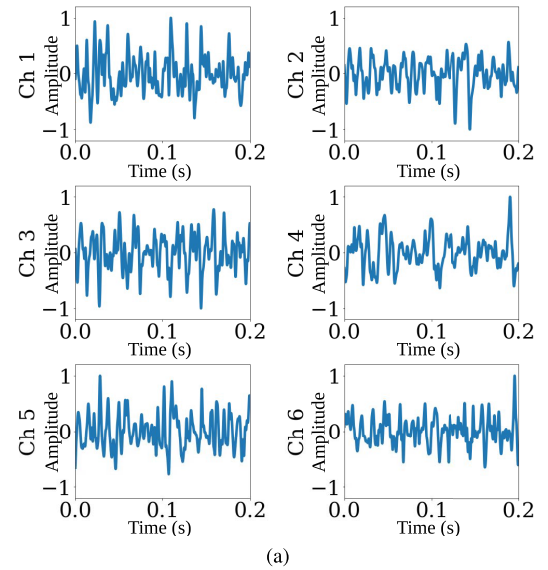


Fig. 3. (a) Windowed sEMG signal with six channels. Mel spectrograms of the windowed signal with the parameter settings of (b) $f_{\max} = 128$, $n_{\text{fft}} = 1024$, and hops = 16 and (c) $f_{\max} = 256$, $n_{\text{fft}} = 1024$, and hops = 128.

each movement and 4400 windowed signals for each subject in a single day. These signals become 30 800 windowed signals for each subject over a period of seven days.

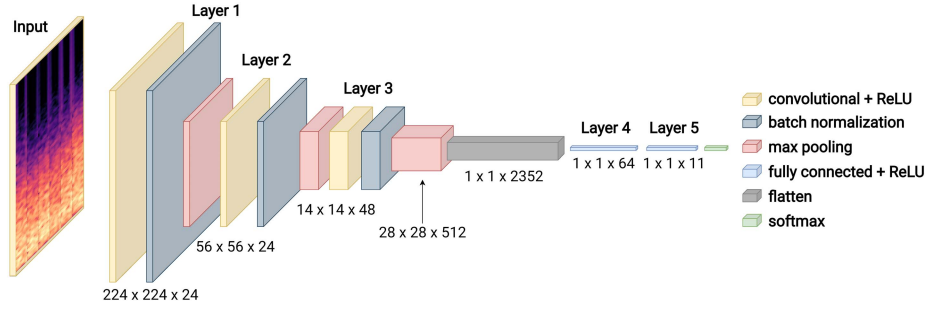


Fig. 4. Architecture of the five-layered proposed CNN.

Since there are six channels (corresponding to the number of electrodes attached to subjects) in each windowed signal as shown in Fig. 3(a), each windowed signal is aggregated vertically and then converted into image as Mel spectrograms for each movement. The Mel spectrograms of the same signal with different parameters' settings are shown in Fig. 3(b) and (c). Here, Mel frequency scales, number of samples per frame in spectrogram, and time scale in the x -axis are represented by $f_{\max}$, n_{fft} , and hops. We are using $f_{\max} = 256$, $n_{\text{fft}} = 1024$, and hops = 128 as depicted in Fig. 3(c). This conversion resulted in 30800 images for each subject.

D. CNN Architecture

The experimental process in this study involves a deep CNN with the max-pooling function. The model is implemented on the Mel spectrogram, and the input size of the feature is 224×224 . The input is denoted as I , where Θ is a parameter of a composite nonlinear function denoted as $G(\cdot|\Theta)$. The operation of convolution can be described as [33]

$$Y = G(I|\Theta) = g_L(\dots g_3(g_2(I|\theta_2)|\theta_3)\theta_L) \quad (3)$$

where g_l is the l th-layer of CNN, and L denotes the total number of layers to be used. In this case, $L = 5$. The parameter for the l th-layer is $\theta_l = [X, b]$. Then the operations in the convolutional layers can be described as

$$Y_l = g_l(I_l|\theta_L) = h(I_l + b * X) \quad (4)$$

where I_l is the input of the l th-layer, X denotes the corresponding filter, $*$ represents the valid convolution, $h(\cdot)$ denotes the pointwise activation function, and b denotes the vector bias term.

The images were fed to CNN with the size of $224 \times 224 \times 3$. A rescaling layer of range (0, 1) is also introduced. In CNN, the Adam optimizer [34] is used, the total number of epochs is 50, the batch size is 32 uniformly, the rectified linear unit (ReLU) activation function [35] is used for the first four layers, and the Softmax activation function [36] is used for the final layer. The general architecture of CNN implementation is depicted in Fig. 4. The parameters of the filters, the strided max-pooling function, and the receptive field are described in the following.

- 1) *Layer 1*: The model's initial layer has 24 filters each with a 9×9 receptive field. A batch normalization layer is used to normalize the input for each mini batch. It is

TABLE III
SUMMARY OF THE PROPOSED MODEL

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 224, 224, 3)	0
conv2d (Conv2D)	(None, 224, 224, 24)	5856
batch_normalization (BatchNormalization)	(None, 224, 224, 24)	96
max_pooling2d (MaxPooling2D)	(None, 56, 56, 24)	0
conv2d_1 (Conv2D)	(None, 56, 56, 48)	28848
batch_normalization_1 (BatchNormalization)	(None, 56, 56, 48)	192
max_pooling2d_1 (MaxPooling2D)	(None, 14, 14, 48)	0
conv2d_2 (Conv2D)	(None, 14, 14, 48)	20784
batch_normalization_2 (BatchNormalization)	(None, 14, 14, 48)	192
max_pooling2d_2 (MaxPooling2D)	(None, 7, 7, 48)	0
flatten (Flatten)	(None, 2352)	0
dense (Dense)	(None, 64)	150592
dense_1 (Dense)	(None, 11)	715
Total params: 207,275		
Trainable params: 207,035		
Non-trainable params: 240		

followed by a 4×4 strided max-pooling function. ReLU is used as the activation function.

- 2) *Layer 2*: The second layer consists of 48 filters with a receptive field of 5×5 . Batch normalization layer is applied. Following that, a 4×4 strided max-pooling is performed using ReLU as the activation function.
- 3) *Layer 3*: The third layer is composed of 48 filters with a receptive field of 3×3 . Padding is the "same" for the feature extraction algorithms. Batch normalization is applied and is followed by a 2×2 maximum pooling using ReLU as the activation function.
- 4) *Layer 4*: The fourth layer is the first fully connected (fc) dense layer, consisting of 64 hidden units with ReLU as its activation function.
- 5) *Layer 5*: The last layer is the second fc dense layer. It is made up of output units equal to the number of classes in the dataset. In the last layer, the Softmax as an activation function is applied.

The summary of the proposed model is given in Table III.

E. Performance Evaluation Metrics

The performance evaluation metrics for multiclass classification such as accuracy, macro weighted precision (MWP), macro weighted recall (MWR), and macro $F1$ -score have been used [37].

TABLE IV
PERFORMANCE OF CNN ON THE ABLE-BODIED (HEALTHY AND NONDISABLED) AND AMPUTEE SUBJECTS

Subjects	Day	Accuracy	Precision	Recall	F1-score	Training Time (s)
Able-Bodied (8 Subjects)	1	99.20%	99.21%	99.20%	99.20%	623.089
	2	99.31%	99.33%	99.32%	99.32%	622.112
	3	99.89%	99.89%	99.89%	99.89%	487.109
	4	99.32%	99.33%	99.32%	99.32%	1040.861
	5	99.89%	99.89%	99.89%	99.89%	1151.958
	6	99.55%	99.55%	99.55%	99.55%	979.991
	7	98.75%	98.77%	98.75%	98.74%	1098.120
Mean±SD		99.42%±0.42%	99.42%±0.15%	99.42%±0.14%	99.42%±0.14%	857.61±11.06%
Amputee (4 Subjects)	1	98.52%	98.54%	98.52%	98.53%	1039.726
	2	95.50%	95.51%	95.53%	95.53%	1048.476
	3	99.50%	99.51%	99.50%	99.50%	1024.135
	4	98.41%	98.43%	98.41%	98.41%	989.428
	5	99.80%	99.81%	99.82%	98.82%	983.722
	6	97.50%	97.52%	97.50%	97.50%	987.859
	7	96.82%	96.84%	96.82%	96.82%	1223.139
Mean±SD		98.00%±0.58%	98.02%±0.54%	98.00%±0.54%	98.00%±0.54%	1042.355±2.82%

The accuracy for multiclass classification is given as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (5)$$

Precision and recall for binary class classification for a generic class k are given as

$$\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k} \quad (6)$$

$$\text{Recall}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}. \quad (7)$$

While for multiclass classification, MWP and MWR are used and given as

$$\text{MWP} = \frac{\sum_{k=1}^K \text{Precision}_k}{K} \quad (8)$$

$$\text{MWR} = \frac{\sum_{k=1}^K \text{Recall}_k}{K}. \quad (9)$$

The macro $F1$ -score for multiclass classification is given as

$$\text{Macro } F1\text{-score} = 2 \times \left(\frac{\text{MWP} \times \text{MWR}}{\text{MWP}^{-1} + \text{MWR}^{-1}} \right). \quad (10)$$

F. Statistical Tests

For performance comparison, Friedman's tests with a two-way layout were applied. The results of classification are presented as an accuracy with standard deviation. Statistical p -values less than 0.05 were considered significant.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Performance Assessment of Proposed Method

This work carries out two types of analysis strategies as follows.

- 1) *Within-Day Analysis* refers to testing a classifier by a single subject's data in one session.
- 2) *Between-Day Analysis* refers to interdays' testing of classifier performance for a single subject's data in all the seven days. This is further divided into two types: 1) twofold in which one day is chosen for training and another one is chosen for testing and 2) sevenfold in which six days are chosen for training and one day is chosen for testing.

TABLE V

BETWEEN-DAY ANALYSIS (TWOFOOLD)
FOR THE ABLE-BODIED (HEALTHY AND NONDISABLED) SUBJECTS

		Test						
	Days	D1	D2	D3	D4	D5	D6	D7
Train	D1	-	75.63	62.72	54.84	54.81	50.61	82.43
	D2	83.15	-	65.13	52.22	72.63	73.90	80.86
	D3	55.34	68.84	-	73.00	81.38	79.02	52.63
	D4	49.63	54.27	68.59	-	67.99	77.25	47.90
	D5	53.75	66.47	70.84	69.68	-	69.27	54.02
	D6	55.11	61.90	63.61	66.65	64.81	-	57.43
	D7	80.61	74.20	69.86	65.75	66.63	71.40	-

1) *Within-Day Analysis*: The training and testing are done on each day for each subject. The data are split into 80–20 format, where 80% data are used for training and 20% data are used for validation. This is done for seven days on each subject. Classification accuracy along with other performance metrics are computed for each day and averaged for respective subjects. For this study, the sEMG data from both healthy and amputee subjects are grouped into two separate datasets. Using CNN, both the sets of data are classified individually. Table IV shows the averaged accuracy, MWP, MWR, and macro $F1$ -score for the combined able-bodied (healthy and nondisabled) and amputee subjects over seven days. The proposed model achieved an accuracy of 99.42% ± 0.42% and 98.00% ± 0.58% for the able-bodied (healthy and nondisabled) and amputee subjects, respectively. Fig. 5 shows the confusion matrix of the proposed model for able-bodied (healthy and nondisabled) subjects, and Fig. 6 shows the confusion matrix for the amputee subjects. Here, the true and predicted labels are the actual hand movements and their respective labels predicted by CNN, respectively.

2) *Between-Day Analysis*: The twofold and sevenfold validation techniques are used in this study.

For the purpose of twofold validation, each combination of days was analyzed in the same way, with one day being used for training and the other being used for validation. Tables V and VI show the per-day classification accuracy of the proposed model for twofold validation. The mean of all the combinations is calculated to get the final result, and the proposed CNN achieved an accuracy of 65.88% ± 10.1%

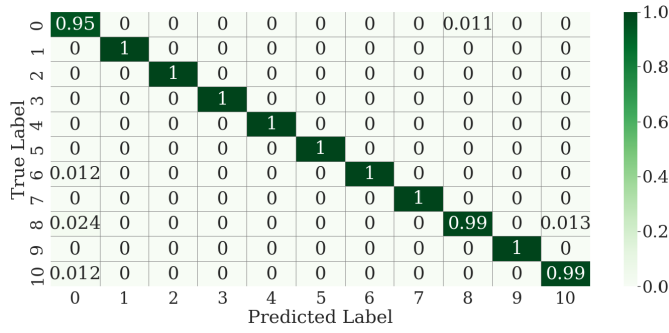


Fig. 5. Confusion matrix of the proposed model for the able-bodied (healthy and nondisabled) subject.

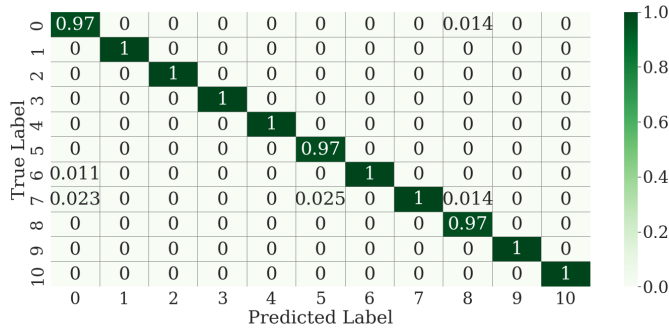


Fig. 6. Confusion matrix of the proposed model for the amputee subject.

TABLE VI
BETWEEN-DAY ANALYSIS (TWO FOLD) FOR
THE AMPUTEE SUBJECTS

		Test						
Train	Days	D1	D2	D3	D4	D5	D6	D7
	D1	-	68.55	67.23	50.84	47.97	44.54	76.51
	D2	68.81	-	54.21	52.22	59.91	54.55	58.53
	D3	40.23	55.56	-	74.35	67.39	70.12	51.29
	D4	48.57	50.71	63.11	-	60.75	57.28	67.37
	D5	52.52	60.45	71.22	61.21	-	63.98	50.12
	D6	58.81	51.90	59.41	60.74	50.12	-	55.72
	D7	77.54	60.77	49.43	65.98	44.25	45.45	-

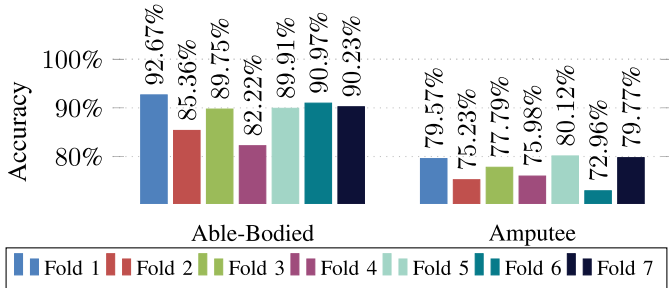


Fig. 7. Performance of the proposed model for the between-day analysis (sevenfold cross validation).

and $58.37\% \pm 9.11\%$ in twofold validation using sEMG data from the able-bodied (healthy and nondisabled) and amputee patients, respectively.

For the sevenfold validation, six days were assigned for training and one day was allotted for validation with seven repetitions, for a total of seven days. Our proposed CNN achieved an average accuracy of $88.73\% \pm 1.43\%$ and $77.35\% \pm 2.72\%$ in sevenfold validation for the able-bodied

(healthy and nondisabled) and amputee subjects, respectively. Fig. 7 shows the performance of the proposed model for both the able-bodied (healthy and nondisabled) and amputee subjects.

B. Comparison and Analysis of Results With Previous and Baseline Methods

This study used a longitudinal dataset collected for seven consecutive days for both the able-bodied (healthy and nondisabled) and amputee subjects. Since this dataset was previously analyzed with SSAEs [32], we are comparing our results with the previous results based on same dataset. CNN showed improved performance than SSAE by achieving $99.42\% \pm 0.37\%$ and $98\% \pm 0.58\%$ mean classification accuracy, while SSAE achieved $96.78\% \pm 6.40\%$ and $86.89\% \pm 11.35\%$ accuracy in the within-day analysis for the able-bodied (healthy and nondisabled) and amputee subjects, respectively. In the between-day (twofold) analysis, CNN performance was $63.46\% \pm 1.96\%$ and $57.78\% \pm 3.61\%$, while SSAE achieved an accuracy of $68.13\% \pm 6.07\%$ [able-bodied (healthy and nondisabled)] and $39.22\% \pm 10.11\%$ (amputee). In the between-day (sevenfold) analysis, CNN achieved the accuracies of $88.73\% \pm 1.43\%$ [able-bodied (healthy and nondisabled)] and $77.35\% \pm 2.72\%$ (amputee), while that of SSAE was $79.57\% \pm 8.56\%$ [able-bodied (healthy and nondisabled)] and $43.67\% \pm 12.15\%$ (amputee). The performance comparison of the proposed model with that of the existing study is shown in Fig. 9.

C. Comparison and Analysis of Results With TL Models

We use the pretrained TL models to the Mel-spectrogram-based datasets of both the subject types for the within-day and between-day analyses. These TL models include AlexNet, VGG16, ResNet18, and MobileNet, as well as DenseNet121 and are widely used in image classification. Table VII shows the results of the within-day analysis for the TL models. The details of each model, including their mean classification accuracy, and their mean training and prediction time, are included in the table, along with the details of our proposed model. The performance of all the mentioned models in terms of accuracy and loss is depicted in Fig. 8. The same TL models are used for the between-day analysis. The results of the analysis are illustrated in Fig. 9.

IV. DISCUSSION

This study investigated the performance of the proposed CNN in short-term and long-term EMG classification. The use longitudinal dataset was recorded over seven days for both the able-bodied (healthy and nondisabled) and amputee subjects. We have experimented and concluded that a Mel spectrogram EMG provides much more information and illustrates better results than a raw EMG signal. However, we used a dataset that comprised a Mel spectrogram composed of six EMG channels. More channels would result in bigger images, which will result in an increase in the computation cost and an increase in the training time. It is also validated that CNN showed improved

TABLE VII
COMPARISON WITH THE PRETRAINED TL MODELS ON THE WITHIN-DAY ANALYSIS

Model	Parameters	Layers	Able-Bodied		Amputee	
			Accuracy $\pm$ SD (%)	Training Time (s)	Accuracy $\pm$ SD (%)	Training Time (s)
Proposed Model	207,275	5	99.42 $\pm$ 0.42	857.61 $\pm$ 11.06%	98.00 $\pm$ 0.58	1042.355 $\pm$ 2.82%
AlexNet	6,848,523	8	94.20 $\pm$ 2.71	1255.519 $\pm$ 11.21%	93.67 $\pm$ 3.59	1396.127 $\pm$ 10.51%
VGG16	138,357,544	16	97.84 $\pm$ 1.42	5898.914 $\pm$ 13.10%	96.36 $\pm$ 2.21	6205.708 $\pm$ 12.35%
ResNet18	11,192,532	18	98.77 $\pm$ 1.12	1778.760 $\pm$ 12.51%	97.79 $\pm$ 1.57	1519.671 $\pm$ 11.86%
MobileNet	3,240,139	28	98.54 $\pm$ 0.99	1926.121 $\pm$ 15.74%	97.31 $\pm$ 0.76	1796.425 $\pm$ 13.51%
DenseNet121	7,037,504	121	98.43 $\pm$ 0.44	5485.165 $\pm$ 11.87%	97.88 $\pm$ 0.52	6645.366 $\pm$ 15.23%

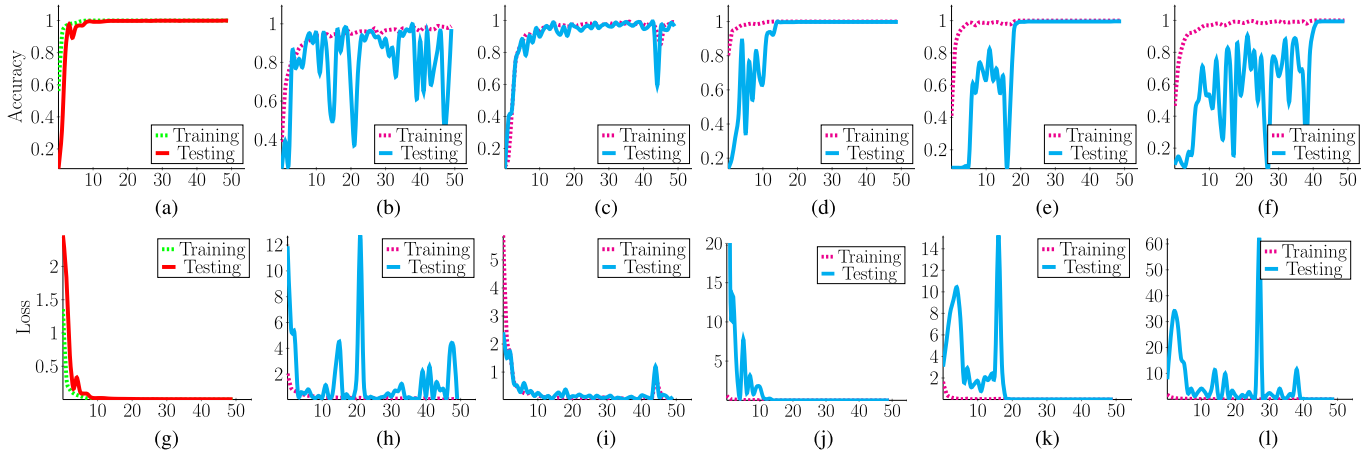


Fig. 8. Performance of the TL models: accuracy for (a) our proposed model, (b) AlexNet, (c) VGG16, (d) ResNet18, (e) MobileNet, and (f) DenseNet121. Loss for (g) our proposed model, (h) AlexNet, (i) VGG16, (j) ResNet18, (k) MobileNet, and (l) DenseNet121.

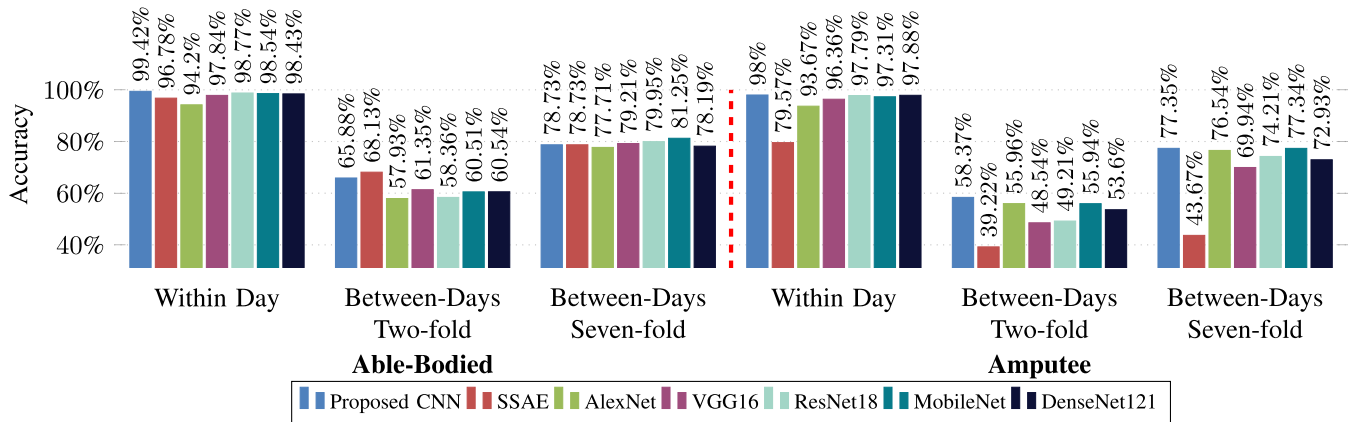


Fig. 9. Comparison of the proposed CNN with other classifiers.

performance over other state-of-the-art approaches, such as SSAE, in terms of accuracy. The results are comparable to those obtained using SSAE; however, SSAE involves the incorporation of handmade features that are similar to those used by the conventional machine learning methods.

The pretrained TL models were also applied to the same image-based dataset. These TL models are based on CNN and have been trained on millions of images with a large number of parameters and are widely used in image classification. The TL models also perform well in sEMG classification, but the computational cost is high. Several investigations indicate that the proposed CNN not only outperforms the TL models but also has a lower computational cost and fewer parameters.

It can be observed that a high number of parameters lead to an irregular training and validation curve. If the training had been terminated earlier, the TL models would not have learned all the characteristics, leading to a worse performance.

In the case of the between-day analysis, the suggested CNN model initially produced average results. This was enhanced when a sevenfold classification was introduced into it. Although the achieved accuracy in the between-days analysis is still lower than the within-day analysis, it is due to the fact that EMG signals are stochastic in nature and its patterns vary largely even within the same session. The proposed model not only significantly addressed this issue for the within-session variability but also improved the

performance in comparison to the previous similar work [32] for the long-term analysis.

One limitation of the study is that both the within- and between-day analyses for both the able-bodied (healthy and nondisabled) and amputees are performed offline. However, future work will be focused on the validation of the proposed model with real-time performance and analyses.

V. CONCLUSION

This research demonstrated that the proposed CNN model showed improved performance over the state-of-the-art SSAE and well-known pretrained TL models. Despite the fact that the nature of EMG is stochastic, however, the within- and between-day investigation showed that the proposed CNN model is capable of improving the performance in long-term classification. This has a substantial impact on reducing the amount of time and effort required for selecting features or signal transformations before classification. The results demonstrate that CNN can considerably increase the effectiveness of the pattern recognition myoelectric control schemes and can extract deep information from EMG data without incorporation of handmade features' extraction.

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